MIL-S-45179B
10 January 1978
SUPERSEDING
MIL-S-45179A
9 December 1966

MILITARY SPECIFICATION

STITCHERS, CEMENTING AND VULCANIZING, AND ROLLERS, HAND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand stitchers and rollers.

1.2 Classification. The stitchers and rollers covered by this specification shall be of the following types, classes, and sizes as specified (see 6.2):

Type I - Rollers

Size 1 - 1-inch diameter by 1-1/2 inches wide (25.4 mm by 38.1 mm)
Size 2 - 1-1/4 inch diameter by 3/8 inch wide (31.7 mm by 9.33 mm)
Size 3 - 1-1/2 inch diameter by 1-1/2 inches wide (38.1 mm by 38.1 mm)
Size 4 - 1-1/2 inch diameter by 2 inches wide (38.1 mm by 50.8 mm)
Size 5 - 2-inch diameter by 2 inches wide (50.8 mm by 50.8 mm)

Type II - Stitchers

Class 1 - Smooth face

Size 1 - 2-inch diameter by 1/32 inch wide (50.8 mm by 0.81 mm)

Class 2 - Serrated face

Size 1 - 2-inch diameter by 1/16 inch wide (50.8 mm by 1.60 mm)

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5120

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116 - Preservation, Methods Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The stitcher or roller to be supplied under requirements of this specification shall be of the latest configurations of the contractor's product that is available on a commercial basis except for details as required by this specification.

3.2 Materials (see 6.3). The materials shall be of the quality normally used for the purpose in commercial practice.

3.3 Construction. The stitchers and rollers shall consist of a steel roller, a wood handle, a steel yoke, an axle, and bearings. The type II, class 2 stitcher shall have not less than 16 teeth per inch (1.6 teeth per mm) of circumference. The configuration shall be similar to the applicable illustrations in figure 1. The diameter tolerance shall be ± 1/16 inch (1.60 mm). The tolerance for the width shall be 1/32 inch (0.81 mm) for the type I rollers and + 1/64 inch (0.04 mm) minus zero for the type II stitchers.

3.3.1 Yoke. The yoke (or fork) shall be made of steel and shall be designed to receive and support both ends of the roller axle.

3.3.2 Handle. The handle shall be made of straight grained, thoroughly seasoned wood, finished by sanding and waxing belt operations or by the application of shellac or varnish. The end of the handle receiving the yoke (or fork) shall be reinforced by a metal ferrule at least 1/2 inch (12.7 mm) in length. The handle shall be at least 4-1/2 inches (114 mm) in length. The diameter of the handle at the large end (next to the ferrule) shall be at least 1-1/8 inches (28.6 mm). The smallest diameter of the handle shall be at least 3/4 inch (19.0 mm).

3.4 Performance.

3.4.1 Impact. The stitcher or roller shall show no evidence of fracture, break, crack or fail to operate efficiently when impact tested as specified in 4.3.1.

3.4.2 Operation. The disc or wheel of the stitcher or roller shall not fail to rotate when tested for operation as specified in 4.3.2.

3.4.3 Strength. The stitcher or roller shall show no evidence of malfunction, fracture, split, warp or breakage after strength testing as specified in 4.3.3.

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3.5 Finish. The stitchers and rollers shall be finished in accordance with the manufacturer's commercial practice.

3.6 Marking. Each stitcher or roller shall be permanently marked with the manufacturer's name or readily identifiable trademark.

3.7 Workmanship. The rollers or stitcher shall be free of sharp edges, burrs, cracks and splits.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all complete stitchers or rollers of one type, class and size offered for inspection at one time. The sample unit shall be one roller or stitcher of one type, class and size.

4.2.2.1 Visual examination. Visual examination of each sample unit shall be made for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified		X
Design	Type, class and size not as specified	X	
Construction and workmanship	Any components missing, defective or loose	X	
	Any sharp edge, burr, crack or split		X
Marking	Missing, incorrect or illegible		X

4.2.2.2 Dimensional examination. Examination shall be made for compliance with dimensional requirements specified. Any dimension not meeting specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. Representative samples from each lot shall be tested in accordance with 4.3.1, 4.3.2 and 4.3.3. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units. Any evidence of failure or deformation resulting from these tests shall be cause for rejection.

4.2.3 Packaging inspection. An examination will be made to determine that preservation-packaging, packing and marking as required by section 5 are complied with. Defects will be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery and need not be sealed. Examination for the closure defects listed below shall be on shipping containers fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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TABLE II. Packaging inspection

Examine	Defect
Preservation	Preservative improperly applied or missing.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps; inadequate stapling, loose strapping; improper taping, bulging or distortion of containers.
Markings	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.
Weight	Net weight exceeds requirements.

4.3 Test.

4.3.1 Impact test. To determine compliance with 3.4.1, the stitcher or roller shall be rolled off a platform 36 inches (914 mm) high and allowed to fall onto a concrete floor not less than five times.

4.3.2 Operation test. The stitcher or roller shall be pushed along a rubberized smooth surface with the roller supporting the weight of the tool alone to determine compliance with 3.4.2.

4.3.3 Strength. Compliance with 3.4.3 shall be determined by rolling the stitcher or roller over the rubber covered platform of a weighing scale with force enough to indicate a scale reading of 40 to 50 pounds (18.1 to 22.7 kg) for a period of ten minutes.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. The metal parts of each stitcher or roller shall be preserved in accordance with method I of MIL-P-116, using type P-9 or P-10 preservative.

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5.1.1.2 Intermediate packaging. Twelve stitchers or rollers of one type, class and size only, shall be intermediate packaged in a fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 125 of PPP-B-636. Box closure shall be in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3.

5.1.2 Level C. Stitchers and rollers shall be preserved-packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A packing. Stitchers or rollers of one type, class and size only, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3. The weight of contents of each shipping container shall not exceed 65 pounds (29.6 kg).

5.2.2 Level B packing. Stitchers or rollers of one type, class and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3. The weight of contents of each shipping container shall not exceed 65 pounds (29.6 kg).

5.2.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3.

5.2.3 Level C packing. Stitchers and rollers, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The hand stitchers and rollers covered by this specification are intended for use in fabrication and repair of coated fabric items and rubber tire and tube repair and rebuild operation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class and size required (see 1.2).
(c) Selection of applicable level of preservation-packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant fiberboard is required for level B pack (see 5.2.2.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.4 Supersession data. Deletion of items no longer required are reflected in comparison with the prrvious revision as follows:

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Deleted	Type I, class 1, sizes 2,3 and 4

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activity:

Army - ME

Preparing activity:

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MIL-S-45179B, Stitchers, Cementing and Vulcanizing, And Rollers, Hand

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